

Solution Description

1 Datasets Used

Logic Dataset (Merged & Augmented):

- **Name/Source:** EXACT Logic Merged & Augmented Set (`logic_merged_valid_augmented.json`). A merge of EXACT 2026 Official Logic-Based Educational Queries (`logic_based.json`, 411 records) and the FOLIO dataset, augmented via quantifier variable canonicalization, entity anonymization, multi-premise permutation, hard negative generation, and synthetic multi-hop logic reasoning problems.
- **Samples:** 1,789 total samples (1,715 train, 74 validation).
- **Sample Entry:**
Input (NL): "If a student is enrolled in a science program and has passed Chemistry 101, they can enroll in Organic Chemistry." → *Question:* "Based on the premises, can we conclude that Minh can enroll in Organic Chemistry?"
Output (FOL): $\forall x(\text{science_program}(x) \wedge \text{passed}(x, \text{chem101}) \rightarrow \text{can_enroll}(x, \text{organicchem}))$
Output (Ans & Explanation): "Yes" → "Minh is enrolled in a science program and passed Chemistry 101, so the premise implies he can enroll..."

Physics Dataset (Preprocessed & Distilled):

- **Name/Source:** EXACT Physics Dataset (`physics.csv`) and EXACT Physics Distillation Set (`physics_distillation.json`). Preprocessed from EXACT 2026 Official Physics problems. The distillation set was generated using teacher models: Google Gemini (Gemini 2.5/3.0/3.5 Flash/Lite variants) and Qwen/Qwen3-235B-A22B-Instruct-2507 (via Together AI) as a fallback.
- **Samples:** 1,352 preprocessed official records, and 1,371 synthetic distilled samples.
- **Sample Entry:**
Input (Question): "Calculate the energy stored in capacitor C when $C = 100\ \mu\text{F}$ and $U = 30\ \text{V}$."
Output (CoT): Step 1: Identify given C and U . Step 2: Recall $E = 0.5 \cdot C \cdot U^2$. Step 3: Convert units...
Output (Answer & Unit): Answer: "0.045", Unit: "J"

2 Approach and Method

Our system uses a standard orchestration layer routing incoming queries to specialized domain pipelines:

Routing & Domain Classification: Queries are classified by type (`type1` logic or `type2` physics) using the `fol_router` adapter. Physics queries are further routed into domain categories (e.g., `electrostatic_force`, `circuit_power`).

Logical Reasoning Pipeline (`type1`): (1) *Glossary-Constrained Translation:* Compiles NL premises and questions into FOL by pre-building a unified JSON glossary of predicates and constants. (2) *FOL Normalization & Repair Loop:* Standardizes quantifiers and connectives, using an LLM-based repair agent to resolve syntax/casing errors. (3) *Z3 SMT Solver:* Mathematically verifies logical entailment. (4) *Neurosymbolic Proof-Guided CoT:* Extracts the minimal Z3 unsat core (if entailed) to guide the LLM in generating a clear, hallucination-free explanation.

Physics Solving Pipeline (`type2`): (1) *Reasoning Policy Assembly:* Dynamically retrieves domain-specific reasoning policies and few-shots based on domain classification. (2) *LLM Code-Gen Solver:* Generates step-by-step reasoning and a Python calculation script. (3) *Sandboxed Execution:* Runs the code in a secure namespace utilizing `sympy` and `pint` for precise unit/circuit calculations. (4) *Self-Correction Loop:* Feeds execution tracebacks or formatting errors back to the LLM to auto-correct the code (max 2 retries).

3 Model Size Calculation

Our pipeline is powered by a single base model with dynamic task-specific adapters loaded and swapped on demand:

Model Component	Total Parameters	Active Parameters
Base LLM (Qwen/Qwen3-8B)	8.0B	8.0B
<code>fol_router</code> Adapter (Rank-64 LoRA)	$\approx 0.35\text{B}$	$\approx 0.35\text{B}$ (When active)
<code>physics</code> Adapter (Rank-64 LoRA)	$\approx 0.35\text{B}$	$\approx 0.35\text{B}$ (When active)
Runtime Peak Footprint	8.7B (Loaded)	8.35B (Class-compliant)

Verification: Both Rank-64 LoRA adapters target all projection modules. The API server loads the base model and both adapters into memory, utilizing PEFT’s dynamic adapter swapping (`set_adapter`). Since only one adapter is active during any single inference forward pass, the active parameters are limited to $8.0\text{B} + 0.35\text{B} = 8.35\text{B}$, fully complying with the 8B-class limit.